

WASP-39b

R-1324

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-39_210527 · created 2026-07-31T19:55:01+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2459361.7578 +/- 0.0072 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.191 +/- 0.054
Transit depth (Rp/Rs)^2	3.63 +/- 2.04 [%]
Orbital Inclination (inc)	89.31 +/- 2.1
Airmass coefficient 1 (a1)	69.45 +/- 37.96
Airmass coefficient 2 (a2)	-2.98 +/- 0.75
Scatter in the residuals of the lightcurve fit is	75.34 %
Best Comparison Star	#1 - [456, 28]
Optimal Aperture	2.85
Optimal Annulus	11.76
Transit Duration (day)	0.124 +/- 0.019

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.191 ± 0.054 vs 0.1457 (deviation 0.72σ)
Duration fit vs published	0.124 d vs 0.12 d (deviation 0.18σ)
Mid-time O–C	7.9 min
Depth SNR	3.0
Residual scatter	75.34 %

Light curve

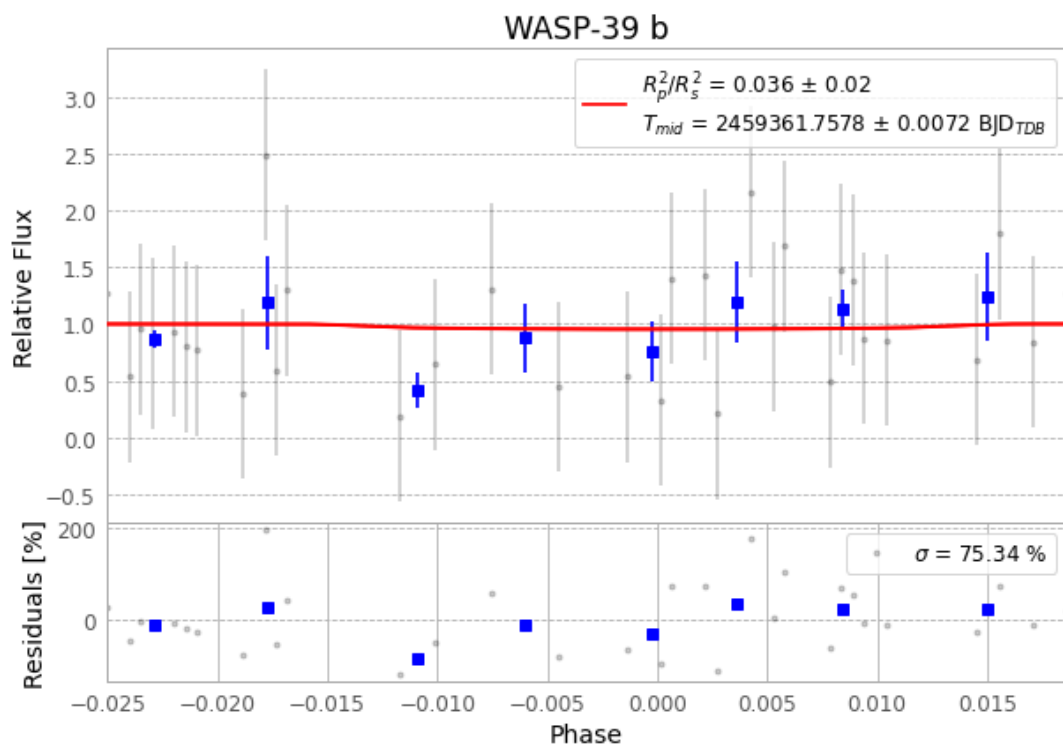


Figure 1 — FinalLightCurve_WASP-39 b_2021-05-26.png (SHA-256 4607b50770a7e96b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [390, 79], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f3fb14a4f926693e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

88 FITS frames (58 MB), WASP-39210527033612.FITS → WASP-39210527075712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-39-210527.log (332a414b8005...), FinalLightCurve_WASP-39 b_2021-05-26.png (4607b50770a7...), FinalLightCurve_WASP-39 b_2021-05-26.csv (62877d6adf3d...)

Compute resources

Wall time 280.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 19:55Z.